Lecture 6: Light as an EM Wave, part 2

- EM Waves in linear media

- Special type of solution: Monochromatic plane waves of fixed polarization (express other solutions as superpositions of these)

- Maxwell's equations → boundary conditions determining reflection and transmission at an interface (Fresnel relations)

Maxwell's Equations in material:

with no free charges or currents

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

Polarization density $\vec{P}$

$$\vec{\nabla} \cdot \vec{P} = -\rho_{bound}$$

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

$$\vec{\nabla} \cdot \vec{D} = \cancelto{0}{\rho_{free}}$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

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$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

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$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

Magnetization density $\vec{M}$

$$\vec{\nabla} \times \vec{M} = \vec{J}_{magnetization}$$

$$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}$$

$$\vec{\nabla} \times \vec{H} = \cancelto{0}{\vec{J}_{free}} + \frac{\partial \vec{D}}{\partial t}$$

Maxwell's Equations in linear media:

with no free charges or currents

$$\vec{\nabla}\cdot\vec{E} = \frac{\rho}{\epsilon_0}$$

Polarization density $\vec{P}$

$$\vec{\nabla}\cdot\vec{P} = -\rho_{bound} \qquad \vec{P} = \chi_e\epsilon_0\vec{E}$$

$$\vec{D} = \epsilon_0\vec{E} + \vec{P} \qquad \vec{E} = \frac{1}{(1+\chi_e)\epsilon_0}\vec{D} = \frac{1}{\kappa_e\epsilon_0}\vec{D} = \frac{1}{\epsilon}\vec{D}$$

$$\vec{\nabla}\cdot\vec{D} = \cancelto{0}{\rho_{free}}$$

$$\vec{\nabla}\cdot\vec{B} = 0 \qquad\qquad \vec{\nabla}\cdot\vec{B} = 0$$

$$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} \qquad\qquad \vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$$

$$\vec{\nabla}\times\vec{B} = \mu_0\vec{J} + \mu_0\epsilon_0\frac{\partial\vec{E}}{\partial t}$$

Magnetization density $\vec{M}$

$$\vec{M} = \chi_m\vec{H}$$

$$\vec{\nabla}\times\vec{M} = \vec{J}_{magnetization}$$

$$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M} \qquad \vec{B} = (1+\chi_m)\mu_0\vec{H} = \kappa_m\mu_0\vec{H} = \mu\vec{H}$$

$$\vec{\nabla}\times\vec{H} = \cancelto{0}{\vec{J}_{free}} + \frac{\partial\vec{D}}{\partial t}$$

Traveling EM wave in linear media:

$$\epsilon = \kappa_e \epsilon_0$$

$$\mu = \kappa_m \mu_0$$

$$\approx \boldsymbol{\mu_0}$$

for us often

$$\vec{E} = \frac{1}{\kappa_e \epsilon_0}\vec{D} = \frac{1}{\epsilon}\vec{D}$$

$$\vec{\nabla} \cdot \vec{D} = \cancelto{0}{\rho_{free}} \qquad \vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{B} = \kappa_m \mu_0 \vec{H} = \mu \vec{H}$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \qquad \vec{\nabla} \times \vec{H} = \cancelto{0}{\vec{J}_{free}} + \frac{\partial \vec{D}}{\partial t}$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\vec{E} = \mu\epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

Wave traveling at speed $c = \frac{1}{\sqrt{\mu\epsilon}} = \frac{1}{\sqrt{\kappa_m \kappa_e}} \frac{1}{\sqrt{\mu_0 \epsilon_0}} \equiv \frac{c_0}{n}$.

n is material's "index of refraction"

Fixed-polarization plane wave:

Consider propagation in x direction $\vec{\boldsymbol{E}}(\vec{r},t) = \vec{\boldsymbol{E}}_0 f(x-ct)$

$$\vec{\nabla}\cdot\vec{D} = 0$$

$$\frac{\partial E_x}{\partial x} + \underbrace{\cancel{\frac{\partial E_y}{\partial y}}}_{0} + \underbrace{\cancel{\frac{\partial E_z}{\partial z}}}_{0} = 0$$

$$\boxed{\vec{\boldsymbol{E}} \perp \textbf{\textit{propagation direction}}}$$

Consider $\vec{\boldsymbol{E}} = \hat{\boldsymbol{y}}\boldsymbol{E_0} f(x-ct)$

$$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} \quad \textit{(Faraday's Law)}$$

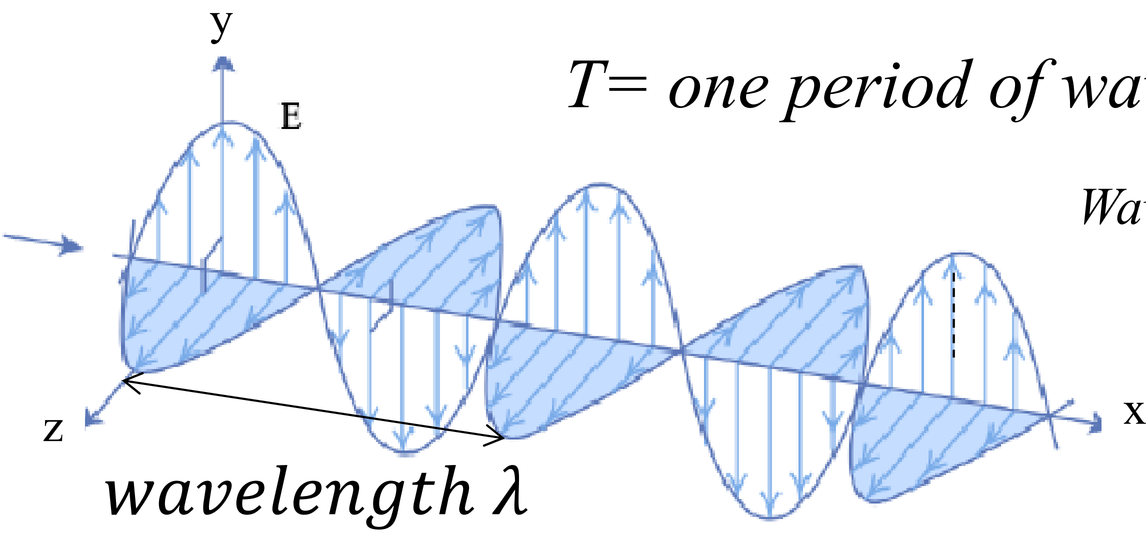
$$\frac{\partial\vec{B}}{\partial t} = -\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & E_0 f(x-ct) & 0 \end{vmatrix}$$

$$\boldsymbol{B} = {}^{\boldsymbol{E}}\!/_{\boldsymbol{c}}$$

$$\vec{\boldsymbol{E}} \perp \vec{\boldsymbol{B}}$$

$\frac{\vec{\boldsymbol{E}}\times\vec{\boldsymbol{B}}}{\boldsymbol{\mu_0}}$ ***gives propagation direction and (energy/time)/area***

Monochromatic Plane Waves:



$wavelength\ \lambda$

$T=$ *one period of wave*

$$\frac{1}{T} = \nu = frequency\ of\ wave$$

Wave number

$$k = \frac{2\pi}{\lambda}$$

Angular frequency

$$\omega = \frac{2\pi}{T} = 2\pi\nu$$

$$\frac{\omega}{k} = \lambda\nu = c$$

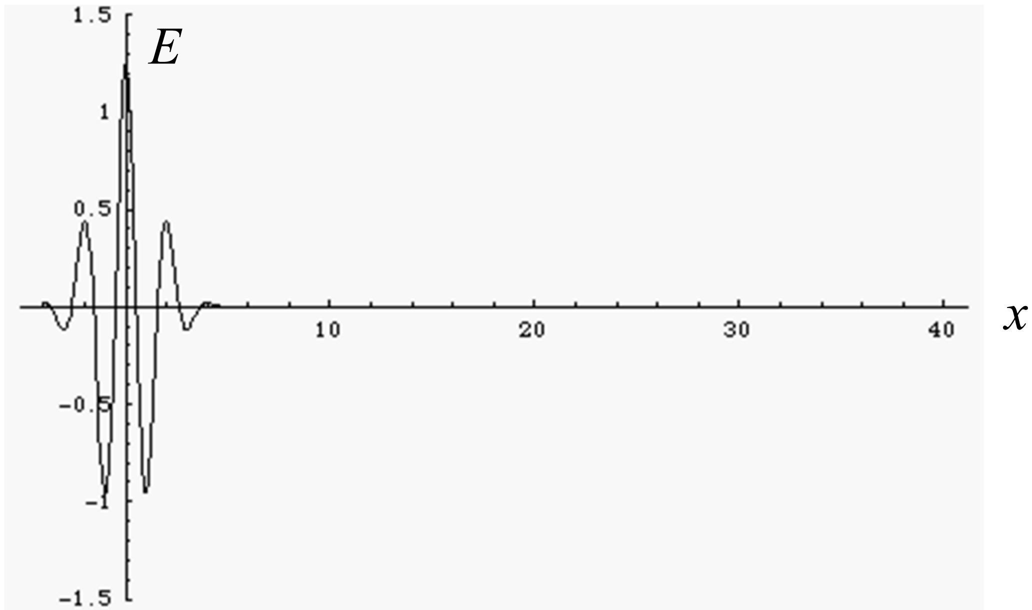
$$\vec{E} = \vec{E}_0 \cos\left(2\pi\left(\frac{x}{\lambda} - \frac{t}{T}\right) + \varphi_0\right)$$

$$= \vec{E}_0 \cos(kx - \omega t + \varphi_0) = \vec{E}_0 \cos\left(k\left(x - \frac{\omega}{k}t\right) + \varphi_0\right)$$

$$= \mathrm{Re}(\vec{E}_0 e^{i(kx-\omega t+\varphi_0)})$$

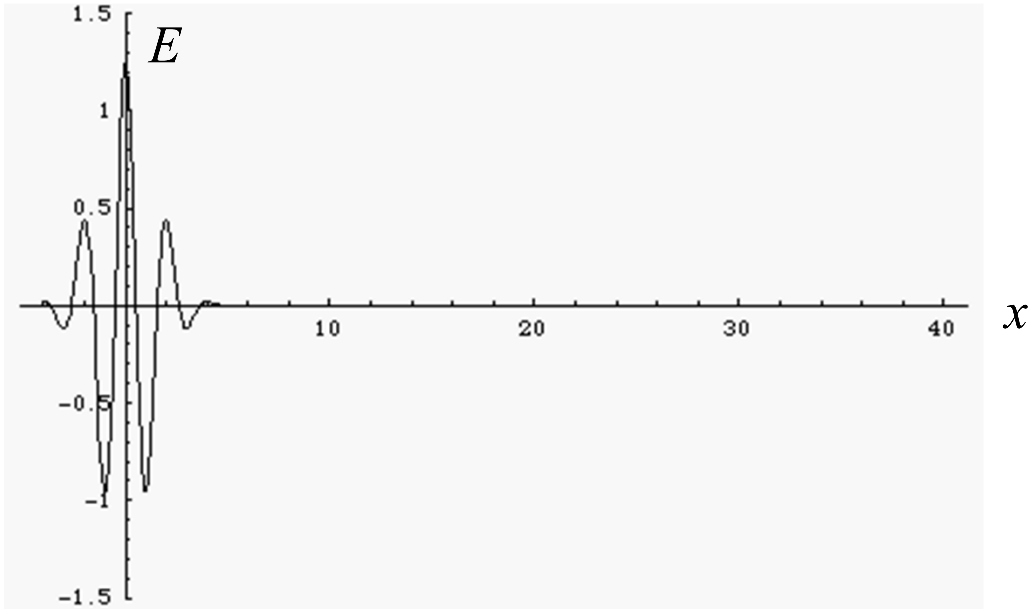
$$= \frac{\vec{E}_0}{2} e^{i(kx-\omega t+\varphi_0)} + \frac{\vec{E}_0}{2} e^{-i(kx-\omega t+\varphi_0)}$$

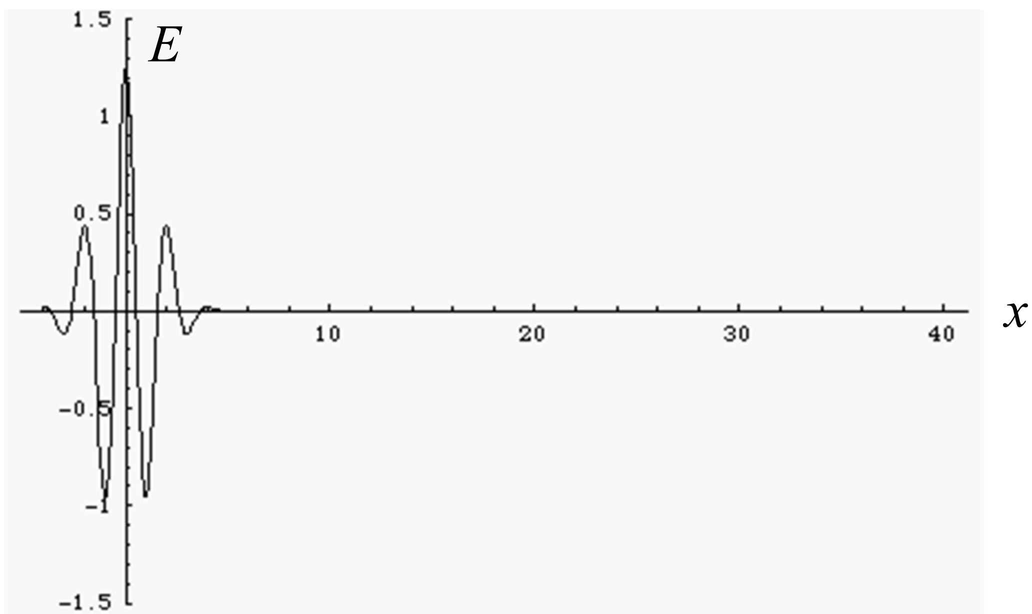
Superposing plane waves of slightly different frequencies: beats, pulses, dispersion



https://www.acs.psu.edu/drussell/demos/dispersion/dispersion.html

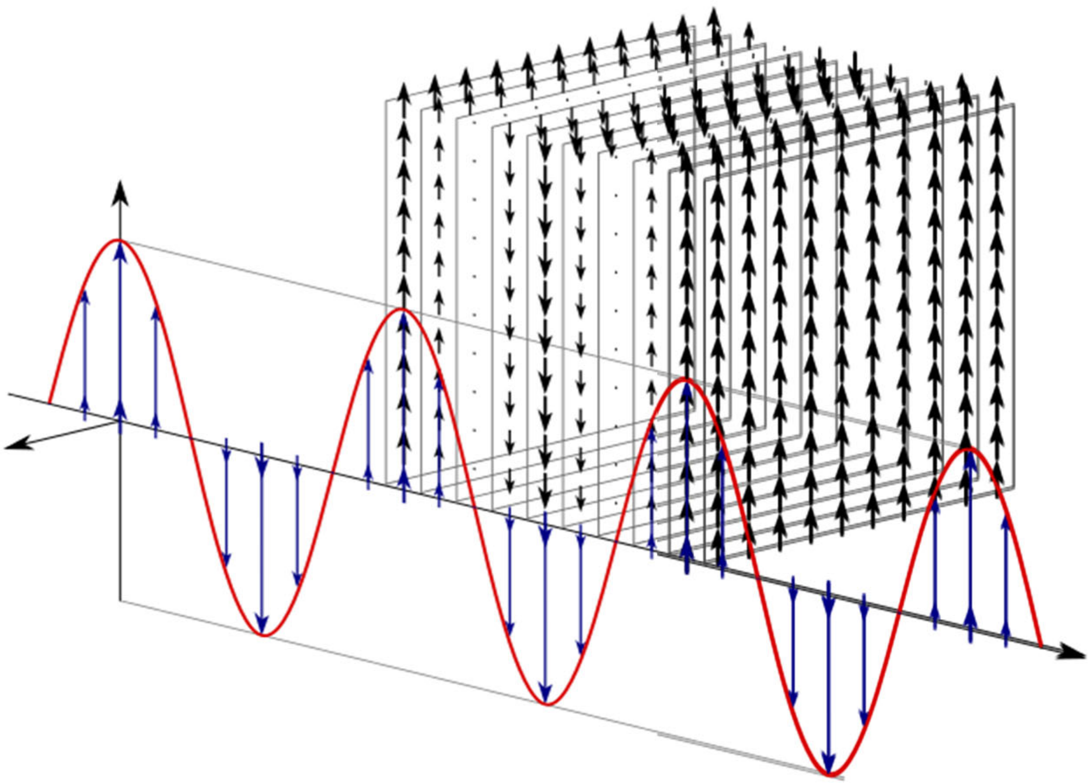
Superposing plane waves of slightly different frequencies: beats, pulses, dispersion





https://www.acs.psu.edu/drussell/demos/dispersion/dispersion.html

Plane waves are not beams!



Superposing plane waves of slightly different propagation directions → beams

Monochromatic Plane Waves:

$$\frac{\omega}{k} = c = \frac{c_0}{n}$$

$$\begin{aligned}\vec{E}(\vec{r},t) &= \mathrm{Re}\left(\vec{E}_0 e^{i(\vec{k}\cdot\vec{r}-\omega t+\varphi_0)}\right)\\ &= \frac{\vec{E}_0 e^{i\varphi_0}}{2} e^{i(\vec{k}\cdot\vec{r}-\omega t)} + \frac{\vec{E}_0 e^{-i\varphi_0}}{2} e^{-i(\vec{k}\cdot\vec{r}-\omega t)}\\ &\equiv \vec{E}_0^{(+)} e^{i(\vec{k}\cdot\vec{r}-\omega t)} + \vec{E}_0^{(-)} e^{-i(\vec{k}\cdot\vec{r}-\omega t)}\\ &= \vec{E}_0^{(+)} e^{i(\vec{k}\cdot\vec{r}-\omega t)} + c.c.\end{aligned}$$

***Intensity** of light is time average over many optical cycles*

$$\langle|\vec{S}|\rangle = \underbrace{\frac{\langle|\vec{E}\times\vec{B}|\rangle}{\mu} = \frac{\left\langle\left|\vec{E}\right|^2\right\rangle}{\mu c} = \frac{\left\langle 2\vec{E}^{(+)}\cdot\vec{E}^{(-)}\right\rangle}{\mu c}}_{\text{Applies very generally}} = \frac{2\left|\vec{E}_0^{(+)}\right|^2}{\mu c}$$

For a monochromatic plane wave

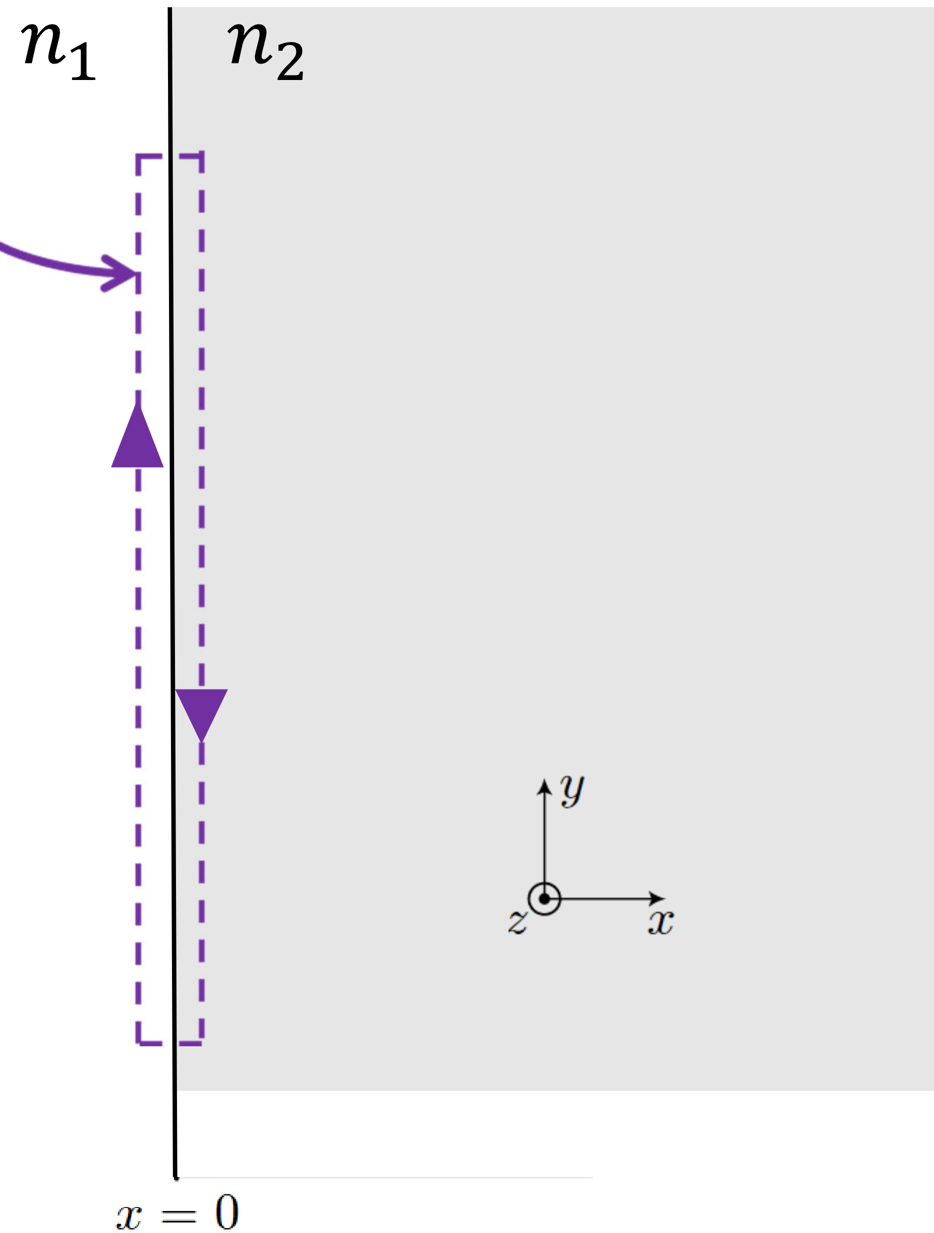
Reflection and transmission at an interface: Maxwell's equations give boundary conditions

$$\oint \vec{E} \cdot d\vec{\ell} = 0$$

around this loop

$$\oint \vec{H} \cdot d\vec{\ell} = 0$$

around this loop



Tangential components of $\vec{E}$ and $\vec{H}$ are continuous across the boundary.